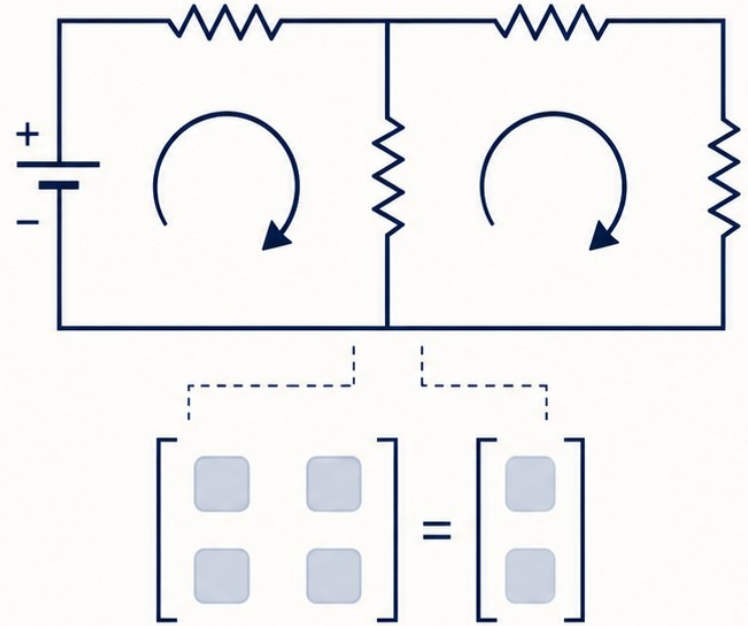


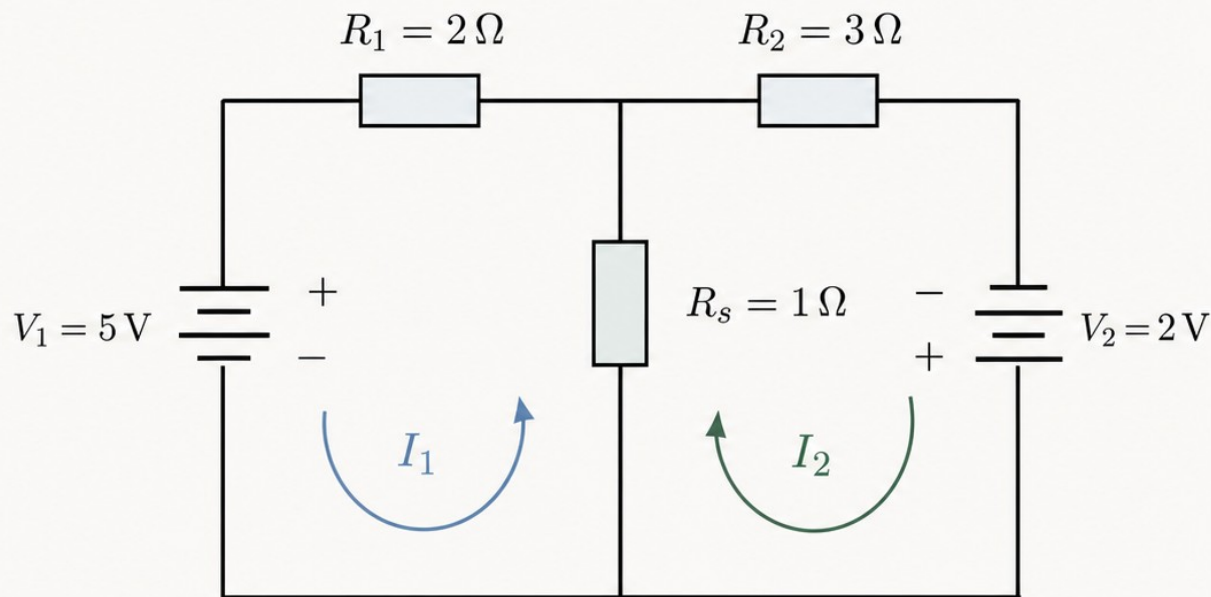
From a circuit sketch to a checkable current

- A two-loop circuit can be written as two equations.
- A 2×2 system stores those equations compactly.
- MATLAB can solve the system.
- The result still needs a physical check.



? Does the current make physical sense?

Predict the current directions before solving



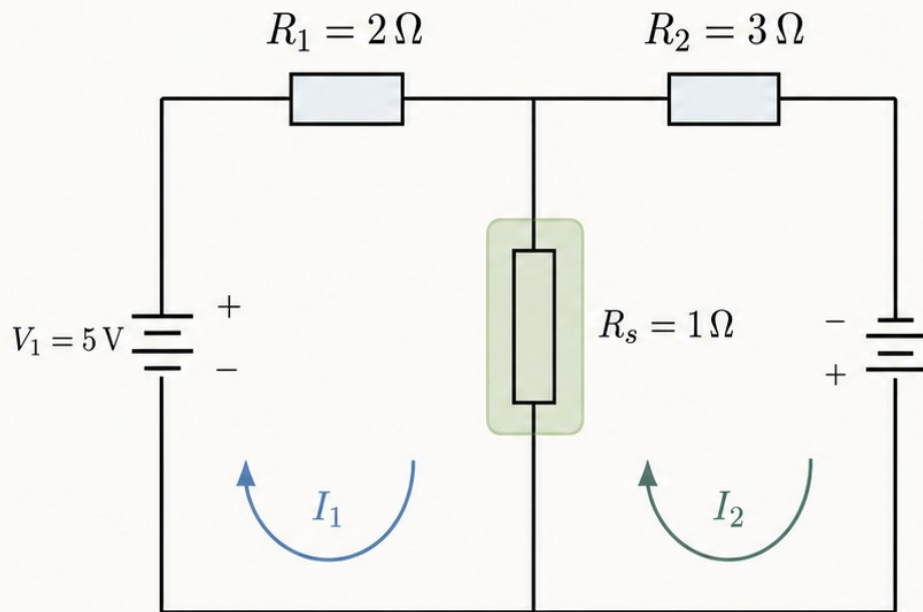
PREDICTION

- ✓ Choose clockwise mesh currents I_1 and I_2 .
- ✓ The chosen directions should give positive currents.
- ✓ The larger source and smaller left private resistance suggest $I_1 > I_2$.
- ✓ A prediction becomes a later check.



Before algebra: which loop do you expect to carry the larger mesh current?

The shared resistor is where the two loop currents meet



I_1



—



I_2



- ✓ The two mesh currents oppose one another in the shared branch.
- ✓ Define the shared direction to follow mesh 1.
- ✓ A negative result means the actual branch direction is opposite to the assigned direction.

$$I_{\text{shared}} = I_1 - I_2$$

positive direction follows mesh 1



What changes if I_2 becomes larger than I_1 ?

Walk each loop in words before writing symbols

LEFT LOOP

follow I_1 through
 R_1 and the
shared resistor

$$\longrightarrow (R_1 + R_s)I_1 - R_s I_2 = V_1$$

shared contribution: $I_1 - I_2$

RIGHT LOOP

follow I_2 through R_2 ;
shared direction
is reversed

$$\longrightarrow -R_s I_1 + (R_2 + R_s)I_2 = V_2$$

shared contribution: $I_2 - I_1$

EVERY TERM IS A VOLTAGE

$$\Omega \times A = V$$

? Say the loop walk aloud before you write the symbols.

Turn each Kirchhoff equation into one matrix row

LEFT LOOP

$$(R_1 + R_s)I_1 - R_s I_2 = V_1$$

Row 1

RIGHT LOOP

$$-R_s I_1 + (R_2 + R_s)I_2 = V_2$$

Row 2

unknown order:
 $\mathbf{x} = [I_1; I_2]$

matching source terms in $\mathbf{b}$

$$\underbrace{\begin{bmatrix} R_1 + R_s & -R_s \\ -R_s & R_2 + R_s \end{bmatrix}}_{A \text{ } (\Omega)} \underbrace{\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}}_{\mathbf{x} \text{ (A)}} = \underbrace{\begin{bmatrix} V_1 \\ V_2 \end{bmatrix}}_{\mathbf{b} \text{ (V)}}$$



ONE EQUATION → ONE ROW

- ✓ Keep the unknown order $\mathbf{x} = [I_1; I_2]$.
- ✓ Row 1 comes from the left-loop equation.
- ✓ Row 2 comes from the right-loop equation.
- ✓ $\mathbf{b}$ stores the matching source terms.



What must stay in the same order?

Keep the units visible in $A \cdot x = b$

$$\underbrace{\begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix}}_{\substack{\mathbf{A} \ (\Omega) \\ \text{resistance} \\ \text{coefficients}}} \underbrace{\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}}_{\substack{\mathbf{x} \ (\text{A}) \\ \text{unknown mesh} \\ \text{currents}}} = \underbrace{\begin{bmatrix} 5 \\ 2 \end{bmatrix}}_{\substack{\mathbf{b} \ (\text{V}) \\ \text{source} \\ \text{rises}}} \quad \Omega \times \text{A} = \text{V}$$



DIMENSIONAL CHECK

$A \cdot x$ has units of volts

$$\Omega \times \text{A} = \text{V}$$



What should each row represent physically?

Audit the matrix before asking MATLAB to solve

which column?

$\mathbf{A} \mathbf{x} = \mathbf{b}$

which row?

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

which sign? which unit?

→ $A_{11} = R_1 + R_s = 3 \Omega$.

→ $A_{12} = -R_s = -1 \Omega$.

→ $A_{21} = -R_s = -1 \Omega$.

→ $A_{22} = R_2 + R_s = 4 \Omega$.

Entry	Value	Physical origin
A_{11}	3Ω	$R_1 + R_s$: resistance multiplying I_1 in the left-loop equation
A_{12}	-1Ω	$-R_s$: effect of I_2 on the shared-resistor drop in the left loop
A_{21}	-1Ω	$-R_s$: effect of I_1 on the shared-resistor drop in the right loop
A_{22}	4Ω	$R_2 + R_s$: resistance multiplying I_2 in the right-loop equation



Read each coefficient back as physics.

Solve the supplied system with MATLAB backslash

```
R1_ohm = 2; R2_ohm = 3; Rs_ohm = 1;  
V1_V = 5; V2_V = 2;  
A_ohm = [R1_ohm+Rs_ohm, -Rs_ohm; ...  
         -Rs_ohm, R2_ohm+Rs_ohm];  
b_V = [V1_V; V2_V];  
x_A = A_ohm\b_V;  
I1_A = x_A(1);  
I2_A = x_A(2);
```

1

Store resistances
and source voltages

2

Build A and b in
the stated order

3

Use $x_A = A_ohm \backslash b_V$

4

Read $x_A(1)$ as I1_A
and $x_A(2)$ as I2_A



EXPECTED OUTPUT

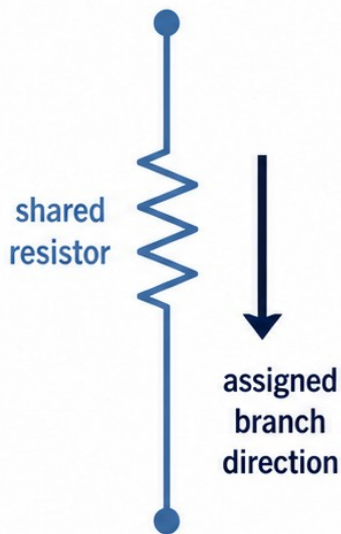
I1_A = 2 A

I2_A = 1 A



What should each row represent physically?

Reconstruct the current in the shared branch



$$I_{\text{shared}} = I_1 - I_2 \quad \text{positive direction follows mesh 1}$$



$$I_1 = 2 \text{ A}, \quad I_2 = 1 \text{ A}$$

$$I_{\text{shared}} = I_1 - I_2 = 1 \text{ A}$$

- The shared resistor carries the difference of the mesh currents.
- $I_{\text{shared_A}} = I_{1_A} - I_{2_A}$.
- The locked solution gives $I_{\text{shared_A}} = 1 \text{ A}$.
- A negative shared value would reverse the assigned branch direction.

RECONSTRUCTED BRANCH CURRENT

$$I_{\text{shared}} = 1 \text{ A}$$



positive: follows
assigned direction



Which direction is positive for the shared branch?

Validate the original equations directly

LOOP 1 RECONSTRUCTED



$$3(2) - 1(1) = 5 \text{ V}$$

LOOP 2 RECONSTRUCTED



$$-1(2) + 4(1) = 2 \text{ V}$$

```
left_reconstructed_V = (R1_ohm + Rs_ohm)*I1_A - Rs_ohm*I2_A;  
right_reconstructed_V = -Rs_ohm*I1_A + (R2_ohm + Rs_ohm)*I2_A;  
validation_difference_V = [left_reconstructed_V - V1_V; ...  
                           right_reconstructed_V - V2_V];  
assert(max(abs(validation_difference_V)) < 1e-12, ...  
       'Direct-substitution validation failed.')
```



DIRECT-SUBSTITUTION CHECK

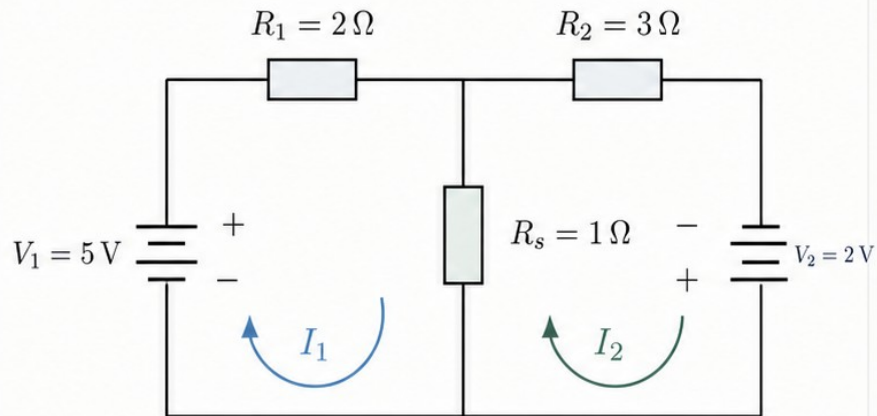
difference vector = [0; 0] V
within the stated tolerance



What do these checks test?

The equations that represent the circuit.

Read the numbers back into the circuit



$$\begin{aligned}I_1 &= 2 \text{ A}, \\I_2 &= 1 \text{ A}, \\I_{\text{shared}} &= 1 \text{ A}\end{aligned}$$

$$\begin{aligned}I_1 &> 0, \\I_2 &> 0, \\I_1 &> I_2\end{aligned}$$



Negative current?

The physical direction is opposite to the assumed arrow — not automatically a failed calculation.



SIGN → DIRECTION

Positive I_1 and I_2 follow the assumed clockwise arrows.



COMPARISON → PREDICTION

$I_1 > I_2$ matches the prediction.



DIFFERENCE → SHARED BRANCH

The shared current follows the mesh-1 direction.

A plausible matrix can still encode the wrong circuit



A solver can still run

```
A_wrong_sign_ohm = [3, 1; ...  
                  -1, 4];  
x_wrong_sign_A = [1.3846; 0.8462];
```

↑
flipped shared-resistor sign

- ✓ Flipping one off-diagonal sign can still produce **finite output**.
- ✓ The wrong matrix changes the **shared-resistor term**.
- ✓ Successful execution is **not proof** of a correct model.
- ✓ This is a **physical/logical defect**, not a MATLAB syntax issue.

finite output

MATLAB syntax
is valid —
but the **physical
model is not**.



Check the original equations

$$A_{\text{ohm}} x_{\text{wrong_sign_A}} - b_V = \begin{bmatrix} -1.6923 \\ 0 \end{bmatrix} \text{ V}$$

correct left-equation mismatch = -1.6923 V

- ✗ The correct equations expose a **-1.6923 V mismatch** in the left equation.
- ✗ The right equation matches (0 V), so the defect is isolated to the left equation.
- ✗ A nonzero mismatch means the model **does not satisfy Kirchhoff's laws**.
- ✗ This is a **physical/logical defect**, not MATLAB syntax.



Repair: trace the shared-resistor term in each Kirchhoff equation.

Working exposure — keep the representation and the physics paired

EQUIVALENT REPRESENTATION

Reordering rows is harmless only when the same reorder is applied to b .

$$\begin{array}{l} Ax = b \\ \text{maximum_reordering_difference_A} = 0 \end{array} \quad \begin{array}{l} \left[\begin{array}{cc} 3 & -1 \\ -1 & 4 \end{array} \right] \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \\ \rightarrow \left[\begin{array}{cc} 4 & -1 \\ -1 & 3 \end{array} \right] \begin{bmatrix} I'_2 \\ I'_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \end{array}$$

$$\begin{array}{l} Ax = b \\ \text{maximum_reordering_difference_A} = 0 \end{array} \quad \begin{array}{l} \left[\begin{array}{cc} 3 & -1 \\ -1 & 4 \end{array} \right] \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \\ \rightarrow \left[\begin{array}{cc} 4 & -1 \\ -1 & 3 \end{array} \right] \begin{bmatrix} I'_2 \\ I'_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \end{array}$$

Same equations, same solution.

PHYSICAL INPUT CHANGE

$$V_2 : 2.0 \text{ V} \rightarrow 2.1 \text{ V}$$

$$\mathbf{x}_{\text{perturbed}} = \begin{bmatrix} 2.0091 \\ 1.0273 \end{bmatrix} \text{ A}$$

$$\Delta \mathbf{x} = \begin{bmatrix} 0.0091 \\ 0.0273 \end{bmatrix} \text{ A}$$

Changing V_2 from 2.0 V to 2.1 V changes both coupled currents.

One source change propagates through coupling.



Equivalent representation preserves the solution; a physical input change propagates through coupling.



OPTIONAL STRETCH

Residual, rank, conditioning, and power balance are optional stretch checks.

Exit ticket — explain the 2x2 solve to a future you



PREDICT

State the unknowns, directions, and units.



MAP

Map one matrix coefficient to its Kirchhoff term.



CHECK

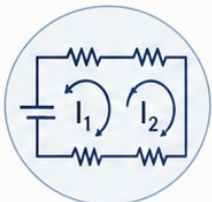
Explain why $A \cdot x$ has units of volts.



INTERPRET

Write one direct-substitution validation and interpret a negative current.

TRANSFER & REFLECTION



Two-loop circuit
(lecture)



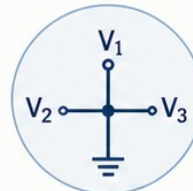
Turn loops
→ equations
→ $A \cdot x = b$



Sensor
(practical)



Model sensor
in circuit and
measure



Node voltage
(analysis)

Same Kirchhoff
ideas in node-
voltage form